

Progress Test 2.B — Elec-S1

2.B · The ohmic conductor and Ohm's law

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Conductance

(1 mark)

A resistor has a conductance $G = 0.25 \text{ mS}$. Give its resistance R in kilohms, exactly.

$R = \frac{1}{G} = \frac{1}{0.25 \times 10^{-3} \text{ S}} = 4.0 \times 10^3 \Omega$	$R = 4.0 \text{ k}\Omega$
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Marking

①
0 1

Marking scheme

① 1 pt $R = 4.0 \text{ k}\Omega$ (G in siemens, $1 \text{ S} = 1 \Omega^{-1}$).

Exercise 2. Ohm's law

(5 marks)

State Ohm's law for a resistor of resistance R , specifying the convention used (draw it). How does the law read in the other convention? State the unit of R (name and symbol) and express it in terms of volts and amperes.

	Receiver convention (arrows opposite): $u = Ri$		Generator convention (arrows in the same direction): $u = -Ri$
$R > 0$ is the resistance, in ohms (Ω): $1 \Omega = 1 \text{ V A}^{-1}$. (Its reciprocal $G = 1/R$ is in siemens.)			

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts Receiver convention drawn (1 pt) with $u = Ri$ (1 pt).
 ② 2 pts Generator convention: $u = -Ri$ (the sign is the point).
 ③ 1 pt Ohm, $\Omega = \text{V A}^{-1}$.

Examiner's tip

“The formula for Ohm’s law changes according to the convention chosen” (notes, remark v). Draw the arrows *before* writing $u = \pm Ri$.

Exercise 3. Resistances in practice

(3 marks)

The typical input resistance of a digital voltmeter is:

- ☐ a few m Ω ☐ a few Ω ☐ a few k Ω ☒ 1 to 10 M Ω

Why: a voltmeter must draw almost no current, so R_V is very large (Tab. 6 of the notes).

To measure, with an ohmmeter, a resistor that is part of a circuit, you must:

- ☐ leave the circuit powered, so that a current flows through the resistor;
☐ switch the power supply off, but leave the resistor connected;
☒ disconnect the power supply and at least one terminal of the resistor;
☐ connect the ohmmeter in series with the resistor.

Why: the ohmmeter injects its own small current; any other source or parallel path would distort the reading. Disconnect, do not merely switch off.

A resistor R carries a current i . The power dissipated by the Joule effect is:

- ☐ Ri ☐ R^2i ☒ Ri^2 ☐ u^2R

Why: $p = ui$ with $u = Ri$ gives $p = Ri^2 = u^2/R$. Check the dimensions: $\Omega \times A^2 = V A = W$.

Exercise 4. Joule effect — a rear-window demister

(5 marks)

A car’s rear-window demister is modelled by an ohmic resistor supplied by the car battery in the DC regime. Determine its resistance R , the current i it carries, then the energy E dissipated as heat in $\Delta t = 10$ min.

$u = 12 \text{ V}$	$p = 72 \text{ W}$	$\Delta t = 10 \text{ min}$
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$$p = \frac{u^2}{R} \Rightarrow R = \frac{u^2}{p} = \frac{(12 \text{ V})^2}{72 \text{ W}} = \frac{144}{72} \Omega \quad R = 2.0 \Omega$$

$$i = \frac{u}{R} = \frac{12 \text{ V}}{2.0 \Omega} \quad i = 6.0 \text{ A} \quad (\text{check: } ui = 12 \times 6.0 = 72 \text{ W})$$

$$E = p \Delta t \text{ with } \Delta t = 10 \times 60 = 600 \text{ s: } E = 72 \text{ W} \times 600 \text{ s}$$

$$E = 4.3 \times 10^4 \text{ J} \quad (43.2 \text{ kJ})$$

Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1
⑤	
0	1

Marking scheme

- ① 1 pt Literal expression $R = u^2/p$, obtained from $p = u^2/R$.
 ② 1 pt $R = 2.0 \Omega$.
 ③ 1 pt $i = 6.0 \text{ A}$ (from u/R or p/u).
 ④ 1 pt $\Delta t = 600 \text{ s}$.
 ⑤ 1 pt $E = 4.3 \times 10^4 \text{ J}$, with its unit.

Your score: / 14 $\rightarrow$ mark out of 20 = score $\times 20/14 =$ / 20